

The surroundings of the germ plasm.

III. The internal temperature of warm-blooded animals (*Mus decumanus*, *M. musculus*, *Myoxus glis*) in artificial climates.

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Differences of average body temperature between adult warm blooded animals dwelling in temperate and in torrid climates respectively have been described only for man. The temperature of various races in the tropics was found to be nearly a degree higher than when they reside in the north temperate zone. (LEFEVRE, 1911, p. 346 ssq.)

The effect on rectal temperature of very short exposure of certain laboratory animals to high temperatures has been described by PEMBREY (1894, 1895). He finds that small changes occur. Thus mice subjected to 30° C. for a period of one half to two hours undergo a raise of 1°. The exposures and observations were not extended to longer periods. SUMNER in a recent article (1911) declares from the evidence of measurements then underway rectal temperature for »almost precisely the same« in mice at - 6° and + 30°.

The following observations were made to find if extended stay at temperatures 10—15° higher or lower than medium produce changes in body temperature. Adult mice, rats and the hibernating animal *Myoxus glis* were used. For a medium temperature, 16° was taken arbitrarily. During the course of the experiments there were indications that animals approaching puberty did not vary to such a degree as observed in adults. A sufficient number of measurements of animals at this stage were accordingly made to settle the point.

Beside the immediate question of the effect of external temperature on the thermal level maintained by the body there was further in mind to find whether germinal and foetal tissues could be modified in temperature by external temperature changes. The relation of such an inquiry to certain problems of evolution is discussed in the introductory article of this series (H. PRZIBRAM, 1911).

Methods.

The »Biologische Versuchsanstalt« afforded rooms maintained continuously very close to 25° and 33° respectively. Other rooms were also used which had average temperatures of 14° and 16° and which fluctuated within a range of 10°. A room was maintained for a time in the winter at an average temperature of 5° and within the limits of 1 and 10° by allowing entry of outside air when necessary.

Most temperature averages discussed are for groups of ten animals measured on four or five days within a period of one or two weeks. The number of measurements are indicated in the tables which follow. Different sets which were compared contained half of one sex and half of the other. Seven individuals out of ten were in their prime and the others of more advanced age. As well as could be judged without weighing, the averages of size were equal for the sets which were compared.

It became soon evident that until the animals were accustomed to the thermometer and to handling, the measurements had little value. Accordingly preliminary readings were taken for three weeks of all animals used and the results discarded. During this time the readings became gradually lower as the animals grew accustomed to the operation. An approximate average of the diurnal temperature changes of rats and mice resulting from the alternation of sleep and activity was obtained by taking half of the readings in the forenoon while the animals were still sluggish and half in the afternoon after they had become active.

Several small sets of animals were used without precautions as to sex age and size. The averages obtained could not be considered as standing for the species in general as all conditions were not represented. The object of these readings was to find the difference in temperature when the sets were changed from one climate to another. For this purpose the averages are obviously trustworthy.

The rats were kept during measurement in a loose wooden tube which allowed sufficient circulation of air. The mice and *Myoxus glis* were held loosely in a towel. Two very small clinical Centigrade thermometers with periods of twenty and thirty seconds respectively were used. The mercury bulb of the former was sufficiently small even for the young mice. During the preliminary readings the temperature was noted every few seconds. A curve was thus obtained that rose abruptly to a correct reading after a length of time corresponding to the period of the thermometer. It then might mount still higher due to the heating resulting from the animal's restless movements. Practice soon made it possible to tell when the true reading was arrived at and also to handle the animal so that the mercury column remained stationary for a considerable time. The thermometer was inserted just deep enough so that its neck was at the level of the anal sphincter muscles. A slight variation in this regard was found to introduce a considerable error. The insertion of the thermometer often caused the rats to discharge faeces. When the rectum was cool, a considerable rise of temperature resulted. The large number of the individual readings that went to make up each average was relied upon to eliminate errors from this source.

Observations.

Table I. Adult rats reared in the temperature at which they were measured.

Average room temperature with difference between room temperatures in brackets	} 16°	[17°]	33°
Average rectal temperature with difference between rectal temperatures in brackets	} 36,2°	[1,0°]	37,2°
Number of individuals	10		10
Number of days measured	4		4

Table I summarizes observations on two sets of ten adult rats in the 16° and 33° rooms who, as well as their parents had been reared at these temperatures. The average rectal temperature in the 16° room was 36,2° and in the 33° room, 37,2°. Between the animals living in the high and in the medium temperatures there was a difference of 1,0°. As previously remarked observations made upon the human temperature for a number of different races living

both in the north temperate and in the torrid zones show a difference of body temperature of 0,9°. The difference of temperature between north temperate and torrid zones is approximately that between the 16° and 33° rooms. The amount of change in body temperature which can be produced by stay in hot surroundings is for man and the rat therefore about equal.

Table II. Adult rats changed from one temperature to another.

Average room temperature with difference between room temperatures in brackets.	5°	[11°]	16°	[9°]	25°	[8°]	33°
Average rectal temperature with difference between rectal temperatures in brackets.	34,4°	[1,8°]	36,1°	[1,7°]	37,8°		
Arrows show temperature from which each set was brought before measurement	← → ←						
Number of individuals in set.	10		5		5		
Number of days measured	4		5		5		
Total period within which measurements were taken	19		7		7		

Table II shows the effect of changing adult rats from one temperature to another. The ten animals in the 5° room had been kept at this temperature for two weeks before measurement. They had previously lived at variable but never extreme temperatures. There was no indication from shivering, restlessness or later sickness that their stay in the 5° room was in any way injurious. Their average temperature before the change was 36,2° and afterwards 34,4°, a difference of 1,8°.

Five individuals that had been reared in the 16° room were changed to the 33° room. Their temperature rose 1,6° to 37,8° during the ensuing week. Five reared in the 33° room and removed to the 16° room fell in average temperature during the first week 1,1° to 36,1°.

Immediately after a change from 16° to 33° the difference in body temperature is 1,7° in contrast to a difference of 1,0° in rats reared at these two temperatures. Thus the thermo-regulatory mechanism can not hold the temperature so low just after a change

to the high temperature as when the animals have been reared at the high temperature. It will be seen later that there is a like difference between young mice reared at and those changed to high temperatures.

The change of adults from 33° to 16° results in a drop of 1,1°. This is only two thirds as great a change as resulted from removal in the opposite direction and differs from the latter also in another regard. Removal to the 33° room caused the temperature to rise 0,6° higher than that of the permanent residents. The opposite change produced a drop only to, not beyond, the temperature of permanent residents at 16°. Such a difference might have been predicted. The powers of thermo-regulation must be taxed at 33° since this temperature is near to the upper limit at which regulation is a possibility. On the other hand since 16° is near to the average temperature of the racial environment, an accurate adjustment would naturally soon take place in animals brought to this temperature.

Table III. Adult mice changed from one temperature to another.

Average room temperature with differences between room temperatures in brackets	}	5°	[11°]	16°	[9°]	25°	[8°]	33°
Average rectal temperature with differences between rectal temperatures in brackets		}	31,2°	[3,0°]	34,2°	[1,1°]	35,3°	[0,4°]
Arrows show temperature from which each set was brought before measurement	}		← → →					
Number of individuals in set.			10		20		10	
Number of days measured		5		5		5		5
Total period within which measurements were taken	}	19		7		5		14

The adult mice of Table III were reared at 16° and as indicated by the arrows part were changed to 5° and part successively to 25° and 33°. The change from 16° to 5° brought about the very considerable fall of 3° in rectal temperature. The animals often shivered but during a months stay in the cold showed no signs of sickness. The ten were together in one nest. It was found at

the termination of the measurements that some of the set might be individuals that three weeks earlier had been removed from a two weeks stay in the 33° room. The adult rats subjected to the change from 16° to 5° underwent a lowering of 1,8° of rectal temperature and did not shiver. It is unfortunate that the possibility of mice having been previously at 33° makes it impossible to say whether the mice actually regulate less well than the rats to low temperatures.

The removal of adult mice from 16° to 25° and later to 33° did not result in a rise commensurate with the fall in the animals exposed to the cold. In the 16° room the temperature was 34,2°, in the 25° room, 35,3°, and in the 33° room, 35,7°. The total raise of rectal temperature was 1,5° for an external change of 17°. The first internal raise was 1,1° and the second, 0,6° for almost equal parts of the external raise. There is not sufficient data to indicate why the first raise is greater than the second. The change of rats from 16° to 25° resulted in a rise of 1,6° which is greater by a half than the mice show. As there is no further evidence at hand that the mice accommodate more accurately than the rats, such a difference between the species can not be considered as satisfactorily established.

Table IV. Adult *Myoxus glis* changed from one temperature to another.

Average room temperature with difference between room temperatures in brackets	14°	[11°]	25°
Average rectal temperature with difference between rectal temperatures in brackets	35,1°	[8°]	35,9°
Arrow shows temperature from which set was brought before measurement	→ ←		
Number of individuals in set	5		5
Number of days measured	10		10
Total period within which measurements were taken . .	10		10

Temperature observations on *Myoxus glis* were taken because it was thought that in as much as the low temperature of winter brings about a hibernating condition in which the body cools off many degrees without injury to the animal that this species would also perhaps be less sensitive than non hibernating animals to high

temperatures and thus that its thermo-regulation would be less accurate. As shown in Table IV the rectal temperature of five adults were taken for ten days. A change was then made to the 25° room and the measurements continued for a second ten days. Before the change the temperature was 35,1°. Afterwards, it rose 0,8° to 35,9°. The mice subjected to a like change experienced a rise of 1,1° and the rats, 1,7°. It is clear therefore that *Myoxus glis* has at high temperatures as accurate or possibly a more accurate temperature regulation than the mice and rats.

Table Va. Rats approaching puberty reared in the temperature at which they were measured.

Average room temperature with difference between room temperatures in brackets	16°	[17°]	33°
Average rectal temperature with difference between rectal temperatures in brackets	37,9°	[0°]	37,9°
Number of individuals in set	10		10
Number of days measured	6		6

Table Vb. Mice approaching puberty reared in the temperature at which they were measured.

Average room temperature with difference between room temperatures in brackets	16°	[17°]	33°
Average rectal temperature with difference between rectal temperatures in brackets	35,9°	[0,2°]	35,7°
Number of individuals in set	3		4
Number of days measured	10		10

In Table Va are measurements of rats just approaching sexual maturity. There were two sets of ten individuals three to four months old of which one was reared in the 16° room and the other in the 33° room. Surprising as it may seem in the light of the difference of 1° found between the body temperatures of older animals reared at the two temperatures, the animals approaching puberty showed the same temperature within a tenth of a degree. The evidence is good as sixty measurements were made of either set and the usual precautions taken. It can not be contended that the 1° raise shown by adults at 33° is due to the presence of many very old individuals

Table Vc. Mice approaching puberty changed from one temperature to another.

Average room temperature with difference between { room temperatures in brackets }	16°	[17°]	33°
Average rectal temperature with difference between { rectal temperatures in brackets }	Set A 34,5°	[1,2°]	35,7°
	Set B 35,9°	[0,7°]	36,6°
Arrows show temperature from which each set was brought before measurements }	Set A ←		
	Set B →		
Number of individuals in set	3—4		3—4
Number of days measured	10		10

which had already lost power of accommodation through weakness. Only two or three of each set showed signs of old age.

A comparison of mice approaching puberty reared at the two temperatures 16° and 33° is summarized in Table Vb. The averages merit less confidence than the preceeding as one set contained but three individuals and the other four. The age of the two sets differed by only a week. Those reared at 33° were slightly smaller and less active. Sex was not noted. The animals reared at 16° had a temperature of 35,9° and those in the heat of 35,7°, that is to say the average in the heat is actually the lower by 0,2°. The greater activity of the animals at medium temperature must have raised the readings a few tenths higher than right for a fair comparison. Experience with measuring the rats and mice for three weeks while they were being tamed showed however that the excess of activity was not great enough to account for more than a few tenths raise. The readings show, that the two sets have temperatures varying by only a small fraction of a degree, if at all.

For rats and mice as well as for a number of other warm-blooded animals, it has been shown that just after birth there is a more or less complete cold blooded condition which in a short time is superceeded by an at least approximately perfect temperature control (EDWARD 1824; PEMBREY 1894, 1895; PEMBREY, GARDEN and WARREN 1894—95). So far as has been previously known after this acquirement of temperature control there is no further change in this function as long as the animal possesses its vigor. We see from the above however that there is a difference in this regard between rats and mice just approaching puberty and those of maturer age.

We may complete the history of thermo-regulation for the rats and mice as follows. The accuracy of thermo-regulation increases to a maximum in a period which includes the time of attaining sexual maturity. It later decreases again while the animals are in full possession of bodily vigour.

The rats approaching puberty average $37,9^{\circ}$ which is higher by $0,7^{\circ}$ than the adults living at the higher temperature and by $1,7^{\circ}$ than the adults living at 16° . Unfortunately we have no data for adult mice reared at 33° . Adult mice brought from 16° to 33° had a temperature of $35,7^{\circ}$ which is that also for the mice approaching puberty. In as much as we have good evidence that adult mice reared at 33° would have considerably lower temperature than those changed from 16° to 33° , it is clear that mice as well as the rats approaching puberty have a higher temperature than the adults reared at 33° . This is an example of the frequently observed rule that the temperature of homothermes tends to vary inversely with the size (see TIGERSTEDT, 1910, p. 39).

In Table Vc is seen the result of changing mice approaching puberty from one temperature to another. The animals used are those whose temperatures before change are given in the last table. The set removed from the 16° to the 33° room underwent a raise of $0,7^{\circ}$ to $35,7^{\circ}$. Those changed in the reverse direction to 16° fell in temperature $1,2^{\circ}$ to $34,5^{\circ}$. There is then a marked modification of temperature after change to the heat although animals at this stage when reared in the heat have the same temperature as when reared at 16° . The average of the changes in the two directions is $0,95^{\circ}$. A similar average of changes produced in adult rats and mice is $1,4^{\circ}$. The animals approaching puberty do not change so much in temperature after removal from one climate to another as do adults. They thus show their superiority to the adults in regulation after change as well as when reared at different temperatures. Perhaps if measurements on the animals approaching puberty for some time longer had been made they might have shown a complete return to the temperature of animals reared in either climate. The transfer to the colder room produced almost twice as great an effect on the rectal temperature as the opposite change. As before stated the animals raised in the hot room were somewhat smaller and less active than the others. It is probable that their temperature changed more than the other set because they were less vigorous and so less resistant rather than that there is

any inherently greater effect produced in changing animals from high to low than from low to high temperatures. Indeed the more reliable averages for adult rats show that for them the reverse is the case.

From the foregoing it is evident that to obtain an average temperature for a homotherm one must take into consideration together with other factors the external temperature and the state of maturity of the animal with especial regard to the period of puberty. If we average the temperatures of animals at puberty and of older animals living at medium temperatures we find the average temperature for the white rat to be $37,0^{\circ}$ and for the white mouse, $34,9^{\circ}$. A variation of two or three tenths in these figures has little significance as changes of this magnitude might easily result from different constructions as to the meaning of average temperature.

Does the temperature of the genital tissue change with that of the environment?

It has been seen that changes of from $0,8^{\circ}$ — 3° have been brought about in rats mice and *Myoxus glis* by a change of 10° to 15° above or below the medium temperature (16°). Between animals at 5° and at 33° there may be a difference of temperature amounting to 4° — 5° . There remains to be considered whether the genital glands and foetus fluctuate in temperature with the rectum.

The testes because of their exposed position may be expected to undergo much greater changes in response to modifications of external temperature than does the rectum unless they be supplied with an especially effective thermo-regulatory mechanism. As a matter of fact we have no evidence for such an especial arrangement. And on the contrary it is a matter of common experience that similar exposed parts in warm-blooded animals as for example hands and feet in man exhibit very considerable fluctuations of temperature. It is accordingly probable that the testes fluctuate as much if not more in temperature than does the rectum.

The ovaries and foetus, in contrast with the testes, are more internally situated than the rectum and would be expected to be more constant in temperature than the rectum. The various observations on the relative temperature of internal organs of warm-blooded animals have established at the most differences of $0,2^{\circ}$

to $0,3^{\circ}$. Comparisons of the relative temperature of more or less internally situated organs as ovaries and rectum are as far as I am aware non-existent. However from the smallness of the above differences between organs there can be little doubt that ovaries and foetus may experience some considerable part of the fluctuations shown by the rectum.

Rats and mice approaching puberty reared at high or medium temperatures would show no difference for ovary and foetus since the rectal temperature remains constant. It may be that the testes even of animals at this age vary in temperature in as much as they are more exposed than the rectum. There is no exemption for animals at any age from fluctuations after they are changed from one temperature to another.

Summary.

1) Adult rats (*Mus decumanus*) reared at 33° have a rectal temperature of $37,2^{\circ}$; at 16° , $36,2^{\circ}$. There is thus a difference of 1° .

2) Rats and mice (*Mus musculus*) approaching puberty show no difference of temperature whether reared at 33° or 16° . The common temperature is for the young rats $37,9^{\circ}$, i. e. $0,7^{\circ}$ higher than for adults reared at 33° . For the young mice it is $35,7^{\circ}$ to $35,9^{\circ}$, also evidently higher than the adults would under like conditions show.

3) Adult rats and mice if changed from 16° to 25° or to 33° undergo a rise of average temperature for the 10 to 20 days following the change of $1,5^{\circ}$ — 2° . At 16° the temperature was $36,1^{\circ}$ for the rats and at 25° , $37,8^{\circ}$. The mice had a temperature of $34,2^{\circ}$ at 16° and of $35,7^{\circ}$ at 33° .

Mice approaching puberty changed from 33° to 16° and vice versa showed temperature changes of about 1° .

4) Adult rats removed from 16° to 5° undergo a fall of $1,8^{\circ}$ to $34,4^{\circ}$ for a period of 19 days. Adult mice some of which may have been earlier at 33° fell 3° after a like change of external temperature. Their temperature was then $31,2^{\circ}$. They were measured during a period of 19 days. They shivered frequently but showed no other signs of ill health during a month's stay which in its entirety averaged little higher than 5° .

5) *Myoxus glis*, a hibernating animal on removal from the 14° room to the 25° room rose in rectal temperature $0,8^{\circ}$ to $35,9^{\circ}$.

The average temperature for the white rat is about 37° and for the white mouse about $34,9^{\circ}$ according to the criterion adopted above. (See last paragraph of observations.)

Zusammenfassung.

1) Erwachsene Ratten (*Mus decumanus*) haben, bei 33° C. aufgezogen, eine rectale Temperatur von $37,2^{\circ}$, bei 16° aufgezogen von $36,2^{\circ}$; es besteht also eine Differenz von 1° .

2) Ratten sowie Mäuse (*Mus musculus*) knapp vor der Geschlechtsreife zeigen keinen Temperaturunterschied, ob sie bei 33° oder bei 16° aufgezogen worden sind. Die gewöhnliche Temperatur für junge Ratten beträgt $37,9^{\circ}$ C., d. h. $0,7^{\circ}$ mehr als jene erwachsener, aufgezogen bei 33° . Für die jungen Mäuse ist sie $35,7^{\circ}$ bis $35,9^{\circ}$, offenbar auch höher, als sie erwachsene unter denselben Bedingungen zeigen würden.

3) Wurden erwachsene Ratten oder Mäuse von 16° nach 25° oder 30° gebracht, so stieg die durchschnittliche Temperatur während der 10 bis 20 unmittelbar auf den Wechsel der äußeren Temperatur folgenden Tage um $1\frac{1}{2}$ bis 2° . Bei 16° war diese Temperatur für die Ratten $36,1^{\circ}$ und bei 25° $37,8^{\circ}$. Die Mäuse hatten bei 16° eine Temperatur von $34,2^{\circ}$, und bei 33° eine solche von $35,7^{\circ}$.

Knapp vor der Geschlechtsreife von 33° nach 16° gebrachte Mäuse und umgekehrt zeigten Temperaturveränderungen von etwa 1° .

4) Wurden erwachsene Ratten von 16° nach 5° gebracht, so sank ihre Rectaltemperatur um $1,8^{\circ}$ auf $34,4^{\circ}$ für einen Zeitraum von 19 Tagen. Erwachsene Mäuse, von denen einige früher bei 33° gehalten worden sein mögen, fielen um 3° nach einer analogen Veränderung der äußeren Temperatur. Ihre Temperatur betrug dann $31,2^{\circ}$. Sie wurden 19 Tage hindurch gemessen. Öfters zitterten sie vor Kälte, zeigten aber keine andern Anzeichen von Unwohlsein während eines Monats, den sie in einer durchschnittlich wenig über 5° betragenden Temperatur zubrachten.

5) Bei *Myoxus glis* (Siebenschläfer), einem winterschlafenden Tiere, stieg die Rectaltemperatur nach Übertragung von einem 14° grädigen Raume in einen 25° grädigen um $0,8^{\circ}$ auf $35,9^{\circ}$.

Die durchschnittliche Temperatur für die weiße Ratte beträgt etwa $37,0^{\circ}$, jene der weißen Maus etwa $34,9^{\circ}$ nach dem oben angenommenen Kriterium (siehe den letzten Paragraphen der Beobachtungen).

(Übersetzt von Hans Przibram.)

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